

VAAGSLI, Norway

WMO: 014340

Lat: 59.7663N

Lon: 7.3650E

Elev: 2697

StdP: 13.32

Time Zone: 1.00 (EUC)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.0	-6.2	-16.4	2.5	-9.7	-11.5	3.3	-5.3	23.4	30.9	19.7	29.9	0.2	230	0.702

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.7	71.1	54.1	67.7	53.0	64.0	52.3	57.4	65.9	55.8	63.4	54.0	60.7	5.6	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
54.3	69.4	58.9	52.3	64.4	57.8	50.4	59.9	56.2	25.8	66.5	24.7	63.2	23.6	60.9	62.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 15.0	(b) 12.1	(c) 10.0	(e) -15.4	(f) 74.7	(g) 5.1	(h) 3.9	(i) -19.1	(j) 77.5	(k) -22.1	(l) 79.8	(m) -25.0	(n) 82.0	(o) -28.7	(p) 84.8		
(a) 15.0	(b) 12.1	(c) 10.0	(e) -15.9	(f) 59.7	(g) 5.1	(h) 2.4	(i) -19.5	(j) 61.5	(k) -22.5	(l) 62.8	(m) -25.3	(n) 64.2	(o) -29.0	(p) 65.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	35.1	19.7	20.2	24.9	32.2	40.3	47.8	53.0	50.6	45.3	36.5	27.9	22.1
	DBStd	14.02	10.53	10.50	9.49	6.76	5.94	4.69	5.28	4.13	4.92	6.05	8.18	11.18
	HDD50	5663	940	834	779	535	306	97	29	43	153	419	662	866
	HDD65	10905	1405	1254	1244	985	765	516	371	448	590	884	1112	1331
	CDD50	234	0	0	0	0	6	31	123	60	13	1	0	0
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH74	16	0	0	0	0	0	0	16	0	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
	WSAvg	4.2	4.5	4.1	4.6	3.9	4.2	4.6	4.1	3.9	4.5	4.1	3.9	4.3
	PrecAvg	43.0	5.5	3.2	3.2	2.1	2.0	2.4	3.2	3.7	4.0	3.9	4.4	5.2
Precipitation	PrecMax	59.3	12.9	10.0	12.2	3.7	4.8	6.6	5.5	7.0	7.3	9.8	8.4	13.1
	PrecMin	26.5	0.2	0.4	0.5	0.7	0.1	0.7	0.6	1.7	1.6	0.6	1.1	0.0
	PrecStd	8.0	3.0	2.4	2.4	0.8	1.2	1.3	1.2	1.5	1.6	2.3	2.1	3.1
	PrecStd	8.0	3.0	2.4	2.4	0.8	1.2	1.3	1.2	1.5	1.6	2.3	2.1	3.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	40.3	43.1	48.9	54.8	68.4	72.0	76.4	70.4	64.9	56.5	46.1	42.1
		MCWB	36.3	37.8	40.2	41.9	48.6	50.8	55.9	54.5	52.3	47.9	42.7	39.3
	2%	DB	37.8	39.2	42.9	50.0	62.4	68.3	72.8	65.7	60.4	51.0	43.2	39.6
		MCWB	34.6	35.0	36.9	40.1	48.9	50.8	55.0	54.3	51.2	45.3	40.6	36.8
	5%	DB	35.7	36.9	39.5	45.8	57.2	64.0	69.9	62.4	56.5	48.0	40.4	37.7
		MCWB	32.8	33.4	35.0	37.9	45.5	49.6	55.0	52.6	49.7	43.5	38.2	35.2
	10%	DB	33.3	34.3	36.8	42.0	52.8	59.7	66.1	59.4	53.7	45.5	38.0	35.5
		MCWB	31.0	31.3	32.6	36.2	43.5	47.7	53.3	51.9	48.9	42.0	35.8	33.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	36.9	38.2	41.4	43.3	52.4	55.4	59.9	58.5	55.5	49.9	43.2	39.4
		MCDB	39.3	41.7	48.9	52.1	64.4	62.9	68.5	64.3	60.4	54.5	45.3	41.5
	2%	WB	35.2	35.7	37.7	40.9	49.6	53.1	58.2	56.4	52.8	46.4	40.8	37.3
		MCDB	37.6	38.4	42.2	48.0	59.6	63.0	68.1	61.7	57.3	50.0	42.9	39.2
	5%	WB	33.2	33.6	35.4	38.8	46.3	51.0	56.9	54.7	51.0	44.2	38.4	35.2
		MCDB	35.3	36.4	39.0	43.9	54.1	59.6	66.4	60.5	55.4	47.1	40.3	37.2
	10%	WB	31.2	31.4	33.0	36.9	44.0	49.2	55.3	52.8	49.2	42.2	36.0	33.3
		MCDB	33.0	34.0	36.2	41.4	52.1	57.7	63.5	57.8	53.1	45.0	37.8	35.2
Mean Daily Temperature Range	5% DB	MDBR	12.8	15.0	16.2	16.0	15.9	17.5	16.7	15.0	13.5	12.0	10.9	12.7
		MCDBR	12.5	13.9	16.5	21.7	25.7	28.3	26.2	21.2	19.4	17.0	12.7	10.6
		MCWBR	10.9	11.0	10.5	13.3	13.5	13.5	12.4	11.6	12.3	11.8	10.8	9.6
	5% WB	MDBR	11.9	12.7	13.8	18.2	22.4	21.6	22.4	18.2	16.8	13.4	11.9	10.8
		MCDBR	10.8	10.5	10.0	11.8	13.0	11.6	12.0	11.2	11.5	10.5	10.5	9.9
		MCWBR	10.8	10.5	10.0	11.8	13.0	11.6	12.0	11.2	11.5	10.5	10.5	9.9
Clear-Sky Solar Irradiance	taub		0.216	0.253	0.284	0.319	0.323	0.319	0.334	0.328	0.301	0.282	0.230	0.191
	taud		2.162	2.260	2.227	2.214	2.331	2.468	2.449	2.458	2.531	2.466	2.222	2.051
	Ebn at Noon		192	246	270	279	287	290	282	276	267	235	183	153
	Edh at Noon		13	21	30	37	35	31	31	29	23	17	12	9
All-Sky Solar Radiation	RadAvg		99	290	698	1278	1491	1574	1437	1126	720	348	128	65
	RadStd		23	71	147	239	263	195	192	98	76	57	25	9

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (58 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page